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NPS/002/016 – Specification for 33kV Cable Joints and Terminations

1. Purpose

The purpose of this document is to detail the technical requirements of 33 kV cable joints and terminations for use on the NEDL / YEDL distribution network.

This document supersedes the following documents, all copies of which should be destroyed.

Ref.	Version	Title

2. Scope

This specification details the technical requirements for 33 kV cable joints and terminations for use on NEDL / YEDL distribution network.

3. Technical Requirements

3.1 Conditions of Operation for Cable Joints.

The following are the general conditions under which cable joints purchased in accordance with this specification are required to operate:

- Nominal system voltages: 33 kV.
- The working voltage of any part of the system does not normally exceed the normal system voltage by more than 10%.
- Nominal system frequency: 50Hz
- The system operates with the neutral point earthed either directly or through a resistance or reactance at one or more points

3.2 33kV Cable Joints and Terminations

Cable joints are required to make a physical and electrical connection between cables of various sizes and types. Existing cables may be:

- Single or three-core paper insulated with copper or aluminium conductors, with either lead or aluminium sheaths with or without armour serving, with or without individual core screening.
- Single or three-core with an extruded insulation with a strippable or bonded screen and copper stranded or solid aluminium core.
- May or may not contain water swellable tapes or powders, a lead or aluminium water block screen or sheath and contain a PVC or PE over sheath.

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Single core joints shall be resin-less with cold applied insulation. Joints containing three core paper cables shall be encapsulated in a filling medium such as resin or resin alternative

All new installations currently utilise single core polymeric cables to BS 7870-4, the range of which is detailed in Appendix 1. All 33kV joints and stop ends will be tested to BS 7888, Cenelec HD 628 and HD 629 or equivalent.

3.3 Terminations

Indoor and outdoor terminations, which shall be supplied in kit form, are to be suitable for polymeric cables. Outdoor terminations will be used where they will be exposed to solar radiation and weathering. The terminations shall be cold applied and contain all the necessary items including mechanical lugs to carry out the connection. The terminations shall be tested to BS 7888, Cenelec HD628 and HD629 or equivalent.

3.4 Insulation Shrouding Boots.

Additional shrouding boots will be cold applied and such that they will fit both straight and angled switchgear bushings. Shrouding boots shall be tested to BS 7888, HD 628 and HD 629.

3.5 Earth Connections

Earth connections will be required to carry 5kA for 2 seconds and should comply with ENATS C93 or equivalent.

3.6 Joints

Joints will be required to connect single core polymeric to single core polymeric in straight and branch joint configurations.

Joints will be required to connect single core polymeric onto three core paper insulated cables, which may have an aluminium or lead sheath.

Joints will be required to connect three core paper cables to three core paper cables, which may have Aluminium or lead sheath.

3.7 Stop Ends

A stop end kit as detailed in BS 7888 is required for existing paper cable to BS 7894 and polymeric cables to BS 7870-4, cable range and size are as detailed in Appendix 1. Cold applied stop ends shall be provided for polymeric cables.

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4 References

4.1 External Documentation

The joints and terminations shall comply with all current versions of the relevant International Standards, British Standard Specifications and all relevant Energy Network Association Technical Specifications (ENATS) current at the time of tendering, except where varied by this standard. In respect the following documents are particularly relevant

Reference	Title	
Energy Network Association Technical Specification		
ENATS-C79	Type Approvals tests for Connectors and Terminations for Aluminium conductors of insulated power cables.	
ENATS-C93	Type Approval tests for mechanical connections to Metallic sheaths of cables	
British Standards		
BS-7888	LV and MV accessories for power cables with rated voltage from 0.6/1kV (Um= 1.2 kV up to and including 20.8/36 kV (Um = 42kV)	
BS-6480	Specification for impregnated paper-insulated lead or lead alloy sheathed electric cables of rated voltage up to and including 33000 V	
BS EN-50180	Bushings above 1 kV up to 36 kV and from 250 A to 3,15 kA for liquid filled transformers	
BS EN-50181	Plug-in type bushings above 1 kV up to 36 kV and from 250 A to 1.25 kA for equipment other than liquid filled transformers	
IEC Technical Specification		
Cenelec HD628	Test methods for accessories for power cables with rated voltages from 3.6/6kV(Um = 7.2 kV) up to and including 20.8/36 (Um = 42kV)	
Cenelec HD629	Test requirements on accessories for use on power cables with rated voltages from 3.6/6kV(Um = 7.2 kV) up to and including 20.8/36 (Um = 42kV	
	Part 1	Cables with extruded insulation.
	Part 2	Cables with unpregnated paper insulation.

4.2 Internal Documentation

Reference	Title
NIL	

5. Definitions

Term	Definition
NIL	

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6. Authority for issue

6.1 Author

I sign to confirm that I have completed and checked this document and I am satisfied with its content and submit it for approval and authorisation.

		Sign	Date
R McMahon	Underground Technical Support	R McMahon	03/04/06

6.2 Policy Sponsor

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
G Bolden	Procurement Manager	G Bolden	23/05/06

6.3 Technical Assurance

I sign to confirm that I am satisfied with all aspects of the content and preparation of this document and submit it for approval and authorisation.

		Sign	Date
D M Gazda	Underground Standards and Assessments Manager	D M Gazda	23/05/06

6.4 DBD Assurance

I sign to confirm that this document has been assured for issue on to the DBD system

		Sign	Date
S Johnson	DBD Administrator	S Johnson	31/03/06

6.5 Approval

Approval is given for the content of this document

		Sign	Date
C Holdsworth	Standards and Assessment Manager	C Holdsworth	23/05/06

6.6 Authorisation

Authorisation is granted for publication of this document

		Sign	Date
C Morgan	Head of Procurement	C Morgan	25/05/06

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Appendix 1 – Typical cable types to be found on the Company's distribution network

33kV

0.25 to 0.4²: Copper conductor, earthed, belted or screened, lead or Aluminium sheath, and PVC or Hessian over sheath

185-300 mm²: Aluminium or Copper Conductor, lead Sheath, and PVC over sheath

185- 400 mm²: Single Core XLPE: Stranded Aluminium core and Copper wire screens

185-500 mm²: Single Core XLPE: Stranded copper conductor and Copper wire screens

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Appendix 2 – Addendum to Supplier Requirements

Work Instructions

The supplier, to a format approved by the company shall write work instructions.

Kitting

All joints including terminations will be supplied in kit form where all the components required including connectors, lugs and work instructions shall be included within the kit.

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Appendix 3 – Logistical Requirements

Storage

To enable NEDL/YEDL to store the product(s) in accordance with the manufacturer's recommendations the Tenderer should provide details of the recommended storage environment, maximum stacking height with respect to each tendered product.

Details should be provided where relevant in respect to the minimum and maximum exposure levels, frequency of exposure and duration of exposure of the packaged item with respect to:

- | | | | |
|---|---------------------|---|-----------------------|
| * | Ambient temperature | * | Atmospheric corrosion |
| * | Humidity | * | impact |
| * | Water | * | Vibration |
| * | Dust | * | Solar radiation |

Packaging and labelling

The Tenderer shall ensure that each item is suitably packaged and protection to maintain the product and packaging as "fit for service" prior to installation taking account of the potential for an outdoor storage environment. All packaging shall be sufficiently durable giving regard to the function, reasonable use and contents of the packaging. Where product packages tendered are made up of sub packages all the sub packages shall unless varied by this specification, be supplied securely packaged together. Where items are provided in bagged/boxed form the material from which the bags are manufactured shall be capable of sustaining the package weight and resisting puncture by the materials within. Tenderer shall submit at the time of tendering the details of the proposed packaging (i.e. materials composition and structure) to be used for each product. Where the Tenderer is unable to provide packaging suitable for outdoor storage then this should be stated at the time of tender.

Palletised goods shall be supplied on standard 1200mm x 1000mm pallets.

Clearly legible, easily identifiable, durable and unambiguous labelling shall be applied to each individual and where relevant multiple package of like products. Where products packages tendered are made up of sub packages each sub packages shall be marked. As a minimum requirement the following shall be included:

- * Manufacturer's trademark or name
- * Supplier's trademark or name
- * Description of item
- * Date of packaging and/or batch number
- * NEDL/YEDL product code
- * Weight

Tenderer shall submit at the time of tendering a sample of the proposed labelling for each product package type.